Who Cares When Care Is Needed? Childcare, Dependent-Adult Care, and Gender in Indian Households

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Abstract

Population averages of care time mix the gender division of care with differences in who lives with someone who needs it. India's Time Use Survey 2024 asks each household whether a member needs special care and records a diary for every member aged six and above. Where a child under six is present, women provide 80 percent of childcare minutes; where a member needs special care, they provide 76 percent of dependent-adult care minutes. Conditional on household characteristics, need is associated with higher adult-care time: by 37.1 minutes a day (SE 1.2) for women and 15.6 (SE 1.0) for men; working-age women's paid work is not detectably different. Men's adult-care time reaches women's average only when no adult woman is present. Family long-term care in India is provided mostly through women's time.

Keywords: care work, eldercare, long-term care, gender, time use, India
JEL codes: J14, J16, J22

1 Introduction

Care for people who cannot fully look after themselves is the core of the unpaid economy (Folbre 1994, 2006; Addati et al. 2018). Its largest component in most countries is childcare, and the time-use literature has studied parental care intensively (Sayer et al. 2004; Craig and Mullan 2011; Folbre et al. 2005). Care of sick, frail and disabled adults has received much less attention, especially outside high-income countries, even though demand for it is rising fastest. India's population aged 60 and above is growing faster than the population as a whole, co-residence with adult children remains the dominant living arrangement for older Indians, and a substantial share of them report difficulty with basic activities of daily living (International Institute for Population Sciences et al. 2020; UNFPA and International Institute for Population Sciences 2023). Public and market provision of long-term care is still developing (UNFPA and International Institute for Population Sciences

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2023; World Health Organization 2015), so most of this care is provided inside the household, by its members, without pay.

This paper asks a question that the existing Indian evidence cannot answer: *when a household needs care, who in it provides that care, and at what cost?* Population-wide comparisons of care time by sex (Hirway and Jose 2011; National Statistical Office 2020) confound two things: the gender division of care within households that need it, and differences between men and women in whether they live in such a household. A man who records no care because no one in his household needs it and a man who records none while his wife nurses his mother both contribute zero to the male average. The same averaging hides the cost of care. Because only a minority of households need adult care on any day, its effect on women's paid work and leisure is diluted when the comparison is made across all households.

India's Time Use Survey 2024 (National Statistics Office 2025) allows the two to be separated. Its household schedule asks whether any member aged five or above needs special care and, if so, whether a caregiver is available among household members. The survey also lists every member and records a 24-hour diary for each member aged six and above, so the time of every potential carer is observed together with the household's need. I use these questions to compare childcare in households with a child under six with care of dependent adults in households with a member who needs special care, and to estimate what that need costs women in paid work, free time and sleep.

Five results stand out. First, in households that need each type of care, women provide 80 percent of household childcare minutes and 76 percent of dependent-adult care minutes; adult care is somewhat less feminized than childcare, but remains predominantly women's work. Women in such households are 21.0 percentage points (SE 1.1) more likely than men in the same kind of household to record any adult care on the diary day. Second, conditional on age, education, consumption, household size and composition, state and day of the week, a household need for special care is associated with higher adult-care time for women, by 37.1 minutes (SE 1.2) and for men by 15.6 minutes (SE 1.0). For women aged 15 to 64 it is associated with 2.0 minutes (SE 3.4) less time in employment and 29.0 minutes (SE 2.9) less free time. Third, the cost depends on the household's own assessment of whether a caregiver is available: women's adult-care time rises more where a caregiver is reported available, consistent with that caregiver usually being a woman. Fourth, 27 percent of all dependent-adult care minutes in need households are provided by people aged 60 and above, so the care profile over the life course is not the one implied by the image of a "sandwich generation" alone. Fifth, men's care rises when no adult woman is present in the household, so men's low care time in mixed households reflects in part the presence of women to whom care is assigned.

The paper contributes to three literatures. For feminist economics of care, it measures care conditional on need, which separates the gender division of care from household demography, and it uses the survey's own assessment of caregiver availability, a direct if coarse measure of the household's care capacity (Folbre 2006; Razavi 2007; Addati et al. 2018). For the economics of informal caregiving,

which comes mostly from Europe and North America (Lilly et al. 2007; Heitmueller 2007; Van Houtven et al. 2013; Bauer and Sousa-Poza 2015), it provides diary-based evidence from a setting where co-residence is the norm, formal care is scarce, and women's attachment to paid work is already weak. For the study of gender and family structure in India, it documents how patrilocal residence and the expectation that daughters-in-law serve their husband's parents translate into minutes of care (Dyson and Moore 1983; Kandiyoti 1988; Lamb 2000; Jayachandran 2015).

The rest of the paper is organized as follows. Section 2 reviews the theory and measurement of care and the evidence on informal adult care. Section 3 sets out a conceptual framework and the questions it implies. Section 4 describes the data and the construction of need, care and carers. Section 5 presents the empirical approach, Section 6 the results, and Section 7 discusses interpretation, limitations and policy.

2 Care, need and gender: theory and evidence

2.1 Care as work, responsibility and relationship

Feminist economics has argued for three decades that care is work, that it is unequally distributed by gender, and that its invisibility in national accounts and economic models has consequences for policy (Folbre 1994, 2001; Himmelweit 1995). Three features of care distinguish it from other unpaid work. First, care is a relationship with a person, not only a set of tasks: its quality depends on attention and continuity, and it involves emotional as well as physical labor (Tronto 1993; Folbre and Nelson 2000; England 2005). Himmelweit (1995) warned that assimilating caring to "work" on the model of paid employment risks losing what is specific to it: much care cannot be separated from the person who gives it or sped up without loss. Second, care involves responsibility and availability. A carer who must remain within earshot of a frail parent is constrained even when not performing a recorded task, which is why time-use researchers distinguish active from passive or supervisory care (Folbre and Bittman 2004; Folbre 2006). Third, as Kittay (1999) argues, those who do dependency work, caring for someone who cannot care for themselves, become dependent in turn: on a partner's earnings, on kin or on the state.

These features explain why care is gendered and why the gendering is persistent. Folbre (1994) analyzes the "structures of constraint" (assets, rules, norms and preferences) that assign the costs of caring for dependants disproportionately to women. Care motivated partly by affection and obligation is difficult to price, and those who provide it may be reluctant to withdraw it even when the terms are unfair, a feature that weakens their bargaining position (Folbre 2001; England 2005). The household is not a unit with a single welfare function but a site of "cooperative conflict" in which perceived contributions and fallback positions shape the division of work (Sen 1990; Agarwal 1997). Women's care work is systematically undervalued in this bargaining because it is unpaid and less visible than earnings.

Care is also organized across institutions. Daly and Lewis (2000) define social care as the activities

and relations involved in meeting the physical and emotional needs of dependent adults and children, and the normative, economic and social frameworks within which they are assigned. Razavi (2007) describes a "care diamond" of family, market, state and community, whose weights vary across countries and over time. Comparative work on Europe shows that care regimes differ in how much they rely on families, with southern European regimes the most family-based (Bettio and Plantenga 2004). In India, the state's main care infrastructure is directed at young children, through the anganwadi system, which is staffed largely by women workers; paid domestic work fills some of the gap for better-off households; and care of adults is expected to come from family members (Palriwala and Neetha 2010). The policy agenda that follows from this literature has been summarized as recognizing, reducing and redistributing unpaid care work (Elson 2017), extended to rewarding and representing care workers (Addati et al. 2018). All three Rs require measurement of who provides care and to whom.

2.2 Measuring care with time-use data

Time-use diaries are the main instrument for measuring unpaid care (Budlender 2010; Charmes 2019). The International Classification of Activities for Time-Use Statistics (ICATUS) 2016 places unpaid caregiving for household members in its own major division, separate from unpaid domestic services, and distinguishes care of children, care of dependent adults and help to non-dependent adult household members (United Nations Statistics Division 2016). This structure makes care of the frail and disabled visible in a way that the older activity lists, which often merged it with housework or "other", did not.

Three measurement problems have been central to the feminist critique of diary data. First, diaries record activities, not responsibilities. Supervisory care and being "on call" are under-recorded unless they are probed, so care time measured as a primary activity is a lower bound on the time constrained by care (Folbre and Bittman 2004; Folbre 2006). Second, care is frequently done alongside other activities, so the primary-activity measure understates its extent and the intensity of the carer's work (Floro 1995). Third, the diary day is a single day. Daily averages correctly estimate the population mean but mix people who never care with people who care every day; the participation rate on a given day understates the share of people who ever provide care, and conditional means overstate its usual intensity (Frazis and Stewart 2012). Time-use surveys in the global South face further problems: high rates of illiteracy require interviewer-recorded diaries, much work is seasonal and informal, and the boundary between production for the market and for the household is blurred (Esquivel et al. 2008; Hirway and Jose 2011).

A less discussed problem matters most for this paper. Comparisons of care by sex across the whole population depend on who lives with a care recipient. Global compilations show women doing about three-quarters of unpaid care work (Charmes 2019; Addati et al. 2018), but they rarely condition on the presence of someone who needs care. For childcare, the age of the youngest child provides a natural conditioning variable, and studies routinely condition on it (Craig and Mullan 2011; Sayer

et al. 2004). For adult care, the recipient is harder to identify, because adults who need help are not defined by age, and most surveys do not ask. Without a measure of need, low male care time cannot be distinguished from low male exposure to need.

2.3 Informal care of adults and labor supply

A large literature, almost entirely from high-income countries, estimates the labor-market cost of caring for an adult. Early work using United States data found that married women's employment responded little to the care of elderly parents (Wolf and Soldo 1994), while co-residential care of a parent reduced women's work hours (Ettner 1996). Lilly et al. (2007) review this evidence and conclude that carers on average are not much less likely to be employed than non-carers, but that intensive carers are more likely to withdraw from work or reduce hours. Studies that address the endogeneity of caring find that treating it as exogenous overstates its employment effect, and that the effect is concentrated among co-resident and intensive carers (Heitmueller 2007). British evidence indicates employment penalties concentrated among those caring for many hours a week (Carmichael and Charles 2003). European panel data show that care reduces employment and hours of work among people aged over 50 (Bolin et al. 2008), with sizeable effects of intensive caregiving on mature women's employment that vary across countries with their care regimes (Crespo and Mira 2014). German evidence shows that effects on women's labor-market attachment persist for years after the onset of care (Schmitz and Westphal 2017). In the United States, informal care has modest effects on women's employment and wages (Van Houtven et al. 2013). Reviews add costs to carers' mental and physical health (Bauer and Sousa-Poza 2015; Coe and Van Houtven 2009), which are larger for women carers, partly because women face more care stressors (Pinquart and Sörensen 2006).

Three features of this literature limit what it says about India. It defines carers by self-report of caregiving, so the counterfactual is a non-carer in any household, and selection into caring is the main identification problem. It studies settings where formal long-term care, pensions and women's employment are far more developed, so the margins of adjustment differ. And it mostly studies care of parents by adult children who live separately, whereas in India co-residence is the norm, and the carer is often a daughter-in-law rather than a daughter.

2.4 Gender, kinship and care over the life course

Gerontology has long documented that family care of frail older people is provided mostly by women, above all by wives and daughters (Stone et al. 1987). Middle-aged women carrying responsibilities to both older parents and their own children became "women in the middle" (Brody 1981) or the "sandwich generation" (Miller 1981). Two qualifications to this picture are important here. First, a substantial share of care for older people is provided by older people themselves, mainly spouses, and in later life men become a large group of carers, mostly for their wives (Arber and Gilbert 1989; Arber and Ginn 1991). The gender gap in care is therefore not constant over the life course. Second,

men's low care time in middle age may reflect the availability of women in the household rather than an absolute absence of male caring. Both qualifications are testable with diary data that cover every member of the household.

In South Asia, obligations to care are shaped by patrilineal and patrilocal kinship. Dyson and Moore (1983) contrast the northern Indian kinship system, with village exogamy and patrilocal residence that separate women from their natal kin, with the southern system, in which women retain more contact with natal kin and more autonomy. Kandiyoti (1988) describes the "patriarchal bargain" of classic patriarchy: a young bride enters her husband's household subordinate to its senior women, and her position improves with age, through sons and through authority over her own daughters-in-law. Das Gupta (1995) shows that women's autonomy in rural Punjab rises over the life course in exactly this way. Ethnographic work on aging in northern India describes the expectation that sons, and in practice their wives, provide *seva* (service and care) to aging parents in return for having been raised (Lamb 2000). Widows' material security depends heavily on co-residence with and support from sons (Chen 2000). These accounts imply that the daughter-in-law is the expected carer of her husband's parents, and that care of the old is, in this sense, part of the marriage contract.

Demographic change is putting pressure on these arrangements. Falling fertility reduces the number of potential carers per older person, migration separates adult children from their parents, and rising life expectancy, longer for women than for men, increases the number of older widows (Rajan et al. 1999; UNFPA and International Institute for Population Sciences 2023). Disability is concentrated at older ages (National Statistical Office 2019; International Institute for Population Sciences et al. 2020). At the same time, India's female labor-force participation is low by international standards and fell for married women between the mid-2000s and the early 2010s (Afridi et al. 2018; Klasen and Pieters 2015; Chatterjee et al. 2018). Explanations emphasize norms, the structure of labor demand, and household responsibilities (Fletcher et al. 2017; Jayachandran 2021). Extended-family structure has been shown to be associated with women's employment (Dhanaraj and Mahambare 2019), but the role of care of adults within that structure has not been measured directly. If care of the old is part of what keeps married women out of paid work, the coming increase in the old-age dependency ratio will make it harder to raise women's employment.

3 Conceptual framework

Consider a household with members $i = 1, \ldots, n$ who allocate their time among paid work, unpaid care, other unpaid work, leisure and sleep. The household's need for care of a dependent member, S_h, raises the total care time the household must supply. How that requirement is divided depends on each member's opportunity cost of time, on bargaining power, and on norms that assign care to particular positions in the household (Folbre 1994; Sen 1990; Agarwal 1997). In a unitary model with comparative advantage, the member with the lowest market wage supplies care, and since women's wages and employment opportunities are lower, women supply most of it. A bargaining

or identity model gives the same prediction for a different reason: care is assigned by gender and position, and departures from the assignment are costly. The models differ in what they predict when the person to whom care is assigned is absent: comparative advantage predicts that the next-lowest-cost member takes over, while strong positional norms predict that care may go unmet or be bought in.

The framework implies four questions that the data can answer.

Q1. Division conditional on need. In households with each need, what share of care time do women provide, and how much of the population gender gap in care is a gap in exposure to need? If women's share of care in need households is similar to their share in the population, low male care is not a matter of exposure.

Q2. Response and cost. How much more care time do men and women record where there is need, and which other uses of time are lower? Paid work is the most-studied margin, but free time and sleep may adjust instead, particularly for women outside the labor force.

Q3. Capacity. The survey asks whether a caregiver is available among household members. A "no" answer can mean that no member is able to care, that the members who could care are occupied with paid work, or that the recipient is cared for from outside the household or not at all. If availability largely reflects the presence of a woman who can be assigned care, women's care time will rise more where a caregiver is reported available. If unavailability reflects strain, the women in those households may bear larger costs per carer.

Q4. Position and life course. Which members of the household supply care, by relationship to the head, sex and age? The kinship literature predicts a central role for daughters-in-law; the gerontological literature predicts a large role for spouses at older ages, and a smaller gender gap among older people. Men's care in households where no adult woman is present tests whether men's low care elsewhere reflects substitution by women.

4 Data and measurement

4.1 The Time Use Survey 2024

The Time Use Survey (TUS) 2024 was conducted by India's National Statistics Office between January and December 2024 (National Statistics Office 2025). It follows the 2019 survey (National Statistical Office 2020), which carried the same household questions on special-care need and caregiver availability and has been used to study women's unpaid work and care (e.g. Neetha 2024); I use the 2024 round because it is the most recent, and compare the main shares with 2019 in Section 7. The survey covers rural and urban India with a stratified multistage design in which first-stage units (villages and urban blocks) are the primary sampling units. Every member of a sampled household aged six and above reports, in a face-to-face interview, the activities of the 24 hours from 4 a.m. on the day before the interview, coded to ICATUS 2016. More than one activity

can be recorded in a time slot, each flagged as major or minor. I use all 10,211,478 activity records, covering 450,457 diaries in 139,487 households and 9,969 first-stage units. Time is aggregated to the person-day from major activities, so that all but 5 of the 450,457 diaries sum to exactly 1,440 minutes (Appendix Table 7). The roster lists 454,192 persons aged six and above; 3,735 of them have no diary records in the file, which leaves the 450,457 diaries, and the 49 diaries of third-gender respondents are excluded from the comparisons of women and men. All estimates use the survey multipliers, and standard errors are clustered on the first-stage unit.

4.2 Need for care

Need for special care is the household's answer to the question "Is there any member (age 5+) who needs special care?" (Block 4, item 17 of the household schedule). The question does not restrict need to adults or to old age, and does not record the type or severity of need or the identity of the recipient. I therefore refer to it as special-care need, not eldercare need. 1.9 percent of households report such a need (2,641 households in the analysis sample). Of these, 66 percent include a member aged 60 or above, so most of the need is likely to concern older people; some recipients will be working-age adults or children aged 5 to 17 with a disability. Among households with a member aged 60 or above, 4.2 percent report a member needing special care.

Caregiver availability is the answer to "Is there any care giver available among household members?" (item 18), asked only when item 17 is "yes". 85 percent of households with a special-care need report a caregiver available. I interpret a "no" as an indicator of constrained care capacity, not as the absence of care: such households may draw on non-resident kin, neighbors or paid help, or care may be provided by members who do not consider themselves available.

Need for childcare is the presence of a child under six. Children under six do not keep diaries; their number is recovered as household size minus the number of diarists. 35 percent of households have a child under six. Children aged five may appear in both need measures, which affects few households.

4.3 Care and its recipients

Childcare is ICATUS 411–419 (physical care, medical care, teaching, talking and reading, playing, minding, and arrangements with schools and childcare providers) plus accompanying own children (442). Dependent-adult care is 421–429 (help with activities of daily living, medical care, administration, emotional support, passive care, and arrangements with care providers) plus accompanying dependent adults (443). Help to non-dependent adult household members (431–439) is excluded from both. Because the recipient of special care may be a child aged 5 to 17, care for whom is coded as childcare, I also report results for all caregiving in ICATUS Division 4.

The person who needs special care also keeps a diary. Including recipients among potential carers would bias the estimates: recipients do little care and little paid work because of the condition that makes them need care, so a household's need would appear to reduce women's employment partly

through the employment of the women who need care. The survey does not identify the recipient. I flag as *probable recipients* members whose usual principal activity status is "not able to work owing to disability" and members who record receiving personal or medical care from others on the diary day (ICATUS 941–949). In 89 percent of households with a special-care need at least one member is flagged. Probable recipients are excluded from all person-level analyzes of carers. The flag is deliberately broad: it also removes some people who visited a doctor, which makes the carer sample slightly more selected towards healthy members but does not affect the household-level shares of care, to which recipients contribute almost nothing.

4.4 Samples

The household-level analysis uses all men and women aged 15 and above in households with non-missing answers to the need questions (388,847 persons). The person-level analysis excludes probable recipients (381,530 persons), and the analysis of costs restricts it further to women and men aged 15 to 64. Transgender respondents are too few for separate estimates and are excluded.

Women's share of household care is computed for each household with positive care of the relevant type as the ratio of women's to all members' minutes, and averaged with household weights. I also report the aggregate ratio of women's to all care minutes. The two differ because the household-level average gives equal weight to households with little and much care.

Table 1: Care provided in households that need it, ages 15+

Need	Outcome	Men	Women
Child under six	Childcare, minutes per day	37.9	124.1
Child under six	Any childcare on diary day	46.7%	82.0%
Child under six	Minutes among those who provide childcare	81.2	151.4
Child under six	Women's share of household childcare		80.5%
Child under six	Women's share of all childcare minutes		78.4%
Member needs special care	Dependent-adult care, minutes per day	15.6	37.1
Member needs special care	Any dependent-adult care on diary day	16.5%	37.5%
Member needs special care	Minutes among those who provide adult care	94.4	98.9
Member needs special care	Women's share of household adult care		75.8%
Member needs special care	Women's share of all adult-care minutes		73.0%
Member needs special care	Women's share of household Division 4 care		77.1%
No member needs special care	Dependent-adult care, minutes per day	0.0	0.0

Notes: Survey-weighted. Persons aged 15+, excluding probable care recipients for person-level rows. Household share: ratio of women's to all members' minutes in households with positive care of that type, averaged with household weights. Aggregate share: women's minutes over all minutes. Division 4: all unpaid caregiving for household members (ICATUS 4xx).

5 Empirical approach

The descriptive comparisons answer Q1. For Q2, I estimate for persons aged 15 and above

$$y_{ih} = \alpha + \beta_1 W_i + \beta_2 S_h + \beta_3 W_i S_h + \beta_4 C_h + \beta_5 W_i C_h + X'_{ih}\gamma + \mu_s + \delta_d + \varepsilon_{ih},$$

where y_{ih} is minutes of an activity on the diary day for person i in household h, W_i indicates a woman, S_h a household member needing special care, C_h a child under six, X_{ih} age band, education, marital status, urban residence, consumption quintile, household size, the number of members aged 60 and above, and whether the diary was reported by proxy, and μ_s and δ_d are state and day-of-week fixed effects. β_2 is men's response to need and $\beta_2 + \beta_3$ women's, which I report with its standard error.

For the costs to women and Q3, I estimate separately for women and men aged 15 to 64

$$y_{ih} = \alpha + \theta^A S_h^A + \theta^N S_h^N + X'_{ih}\gamma + \mu_s + \delta_d + \varepsilon_{ih},$$

where S_h^A and S_h^N indicate a special-care need with and without a caregiver reported available, the reference group is households with no such need, and X_{ih} adds the number of children under six. The pooled version replaces the two indicators with S_h. Outcomes include minutes of dependent-adult care, all caregiving, domestic work, employment, free time and sleep, and two shares: any employment on the diary day and employment in the usual principal status. The last is the most direct labor-supply measure, since it refers to the year before the survey rather than to one day.

These coefficients are conditional associations, not causal effects. Households in which someone needs care are likely to be older, and may be poorer, than other households, and women in them may differ in unobserved ways from other women. The controls absorb the most important observable differences, including the number of older members, which separates the association with need from that with an older household. Two features of the setting limit the most common bias in the caregiving literature, selection into caring: need is measured at the household level, so the comparison is between women in households with and without need rather than between carers and non-carers, and in co-residential households the decision to care is arguably less a separate choice than a consequence of residence. Selection into co-residence remains: families may move a frail parent in with the child best able to care, or a daughter-in-law may withdraw from the labor market before need arises. I return to this in Section 7.

For Q4, I report the distribution of dependent-adult care minutes by the carer's sex and relationship to the head, the age profile of care by sex, and two comparisons: care by spouses of probable recipients, and men's care in need households with and without a non-recipient woman aged 15 or above.

6 Results

6.1 The division of care conditional on need

Table 1 gives the core comparison. In households with a child under six, women aged 15 and above average 124 minutes of childcare a day and men 38. Women are 35.4 percentage points (SE 0.3) more likely than men to record any childcare. In households with a member who needs special care, women average 37.1 minutes of dependent-adult care and men 15.6, a gap of 21.5 minutes (SE 1.4). 37.5 percent of women and 16.5 percent of men in these households record some adult care on the diary day, and those who do spend 99 and 94 minutes respectively. In households with no special-care need, women record 0.0 and men 0.0 minutes of adult care, which confirms that the need question identifies the households where adult care is concentrated.

Measured as women's share of household care minutes, adult care is less feminized than childcare: 76 percent against 80 percent. The aggregate shares, which weight households by the amount of care they provide, are 73 percent and 78 percent. Counting all caregiving for household members, so that care of a disabled child recorded as childcare is included, women provide 77 percent of care in households with a special-care need. Conditioning on need does not narrow the gender gap in care: differences in exposure to need explain little of women's dominance of care.

The degree of feminization differs with the likely recipient. In need households that include a member aged 60 or above, women provide 75 percent of adult-care minutes; in need households with no member aged 60 or above, where the recipient is a working-age adult or an older child, 77 percent. It also differs with care capacity: women provide 76 percent of adult-care minutes where a caregiver is reported available and 77 percent where none is.

6.2 Care time and need

Table 2 estimates how care time differs with need, by gender. Because adult care is almost absent in households without a special-care need (Table 1) and the controls barely shift it, the need coefficients in column 1 are close to the raw means in need households; I checked that this is not a coding artefact (the raw means are 15.60 and 37.10, the coefficients 15.64 and 37.12). In column 1, a household need for special care is associated with 15.6 minutes (SE 1.0) more dependent-adult care for men and an additional 21.5 minutes (SE 1.4) for women, so that the total difference for women is 37.1 minutes (SE 1.2). In column 2, a child under six is associated with 29.1 minutes (SE 0.4) more childcare for men and 110.0 minutes (SE 0.7) for women. Column 3 shows that the special-care difference is larger when all caregiving is counted (46.3 minutes (SE 1.9) for women), so part of the care for special-care recipients is recorded as childcare or help to non-dependent adults.

Columns 4 to 6 show how other activities differ. The need for special care is associated with 7.6 minutes (SE 4.9) less employment for men and 7.0 minutes (SE 3.2) more for women. Women's free time changes by -31.4 minutes (SE 3.0) and their sleep by -6.5 minutes (SE 2.1). Women in need households have less free time and sleep, rather than less paid work.

Table 2: Care need, gender, and time use (minutes per day)

	Adult care (1)	Childcare (2)	All care (3)	Employment (4)	Free time (5)	Sleep (6)
Woman	0.070***	4.26***	4.43***	-236.2***	-11.6***	9.68***
	(0.021)	(0.116)	(0.132)	(1.33)	(0.930)	(0.516)
Member needs special care	15.6***	2.13***	23.8***	-7.61	-12.5***	-4.88**
	(0.952)	(0.693)	(1.48)	(4.89)	(3.66)	(1.96)
Child under six	0.056	29.1***	29.2***	49.5***	-35.2***	2.68***
	(0.039)	(0.438)	(0.453)	(1.56)	(1.21)	(0.684)
Woman x special-care need	21.5***	-2.52**	22.5***	14.7***	-18.9***	-1.66
	(1.44)	(1.22)	(2.08)	(5.60)	(4.04)	(2.18)
Woman x child under six	-0.257***	81.0***	81.0***	-89.7***	2.58**	3.51***
	(0.051)	(0.617)	(0.627)	(1.71)	(1.17)	(0.641)
Observations	381,530	381,530	381,530	381,530	381,530	381,530
R^2	0.19838	0.47668	0.45677	0.39647	0.14257	0.10386
state fixed effects	✓	✓	✓	✓	✓	✓
dow fixed effects	✓	✓	✓	✓	✓	✓

Notes: Persons 15+, excluding probable care recipients. All care: ICATUS Division 4. Controls: age band, education, marital status, urban, consumption quintile, household size, members 60+, proxy report; state and day-of-week fixed effects. Survey-weighted; FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

6.3 The cost to women and the role of caregiver availability

Table 3: Special-care need and the time use of women and men aged 15–64

	Women				Men
Outcome	Any need (1)	Carer available (2)	No carer (3)	Difference (4)	Any need (5)
Dependent-adult care (min)	35.7*** (1.3)	37.9*** (1.4)	20.8*** (2.8)	17.1*** (3.1)	14.3*** (1.0)
All caregiving, Division 4 (min)	49.8*** (1.9)	52.3*** (2.0)	32.8*** (4.8)	19.5*** (5.2)	21.7*** (1.5)
Unpaid domestic work (min)	7.4*** (2.7)	8.3*** (2.9)	1.7 (6.8)	6.6 (7.4)	6.8*** (1.5)
Employment (min)	-2.0 (3.4)	-3.7 (3.6)	9.7 (10.9)	-13.3 (11.5)	-0.8 (5.2)
Any employment on diary day (share)	0.015* (0.009)	0.011 (0.010)	0.043* (0.024)	-0.032 (0.026)	-0.003 (0.009)
Employed, usual status (share)	0.019** (0.009)	0.015 (0.009)	0.041 (0.025)	-0.026 (0.027)	0.010 (0.007)
Free time (min)	-29.0*** (2.9)	-28.6*** (3.2)	-31.9*** (7.4)	3.3 (8.0)	-11.9*** (3.8)
Sleep (min)	-5.2** (2.1)	-5.8** (2.3)	-1.2 (5.9)	-4.6 (6.3)	-6.6*** (2.0)

Notes: Each cell is the coefficient (SE) on the need indicator(s) from a separate survey-weighted regression; reference group: households with no member needing special care. Columns (2) and (3) come from one regression with both indicators; column (4) is (2) minus (3). Probable care recipients excluded. Controls: age band, education, marital status, urban, consumption quintile, household size, members 60+, children under six; state and day-of-week fixed effects. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

Table 3 focuses on working-age women and men. Among women aged 15 to 64, a household member needing special care is associated with 35.7 minutes (SE 1.3) more dependent-adult care, 7.4 minutes (SE 2.7) more domestic work, 2.0 minutes (SE 3.4) less employment, 29.0 minutes (SE 2.9) less free time and 5.2 minutes (SE 2.1) less sleep. The probability of being employed in the usual principal status is 1.9 percentage points (SE 0.9) higher. For men of the same age, the corresponding changes are 14.3 minutes (SE 1.0) more adult care and 0.8 minutes (SE 5.2) less employment.

When probable recipients are kept in the sample, the employment coefficient for women is -13.8 minutes (SE 3.2) rather than -2.0 minutes (SE 3.4). Ignoring the recipient overstates the employment cost of care, because it counts the recipient's own absence from work as a cost to carers. The employment estimates differ in sign across samples: 7.0 minutes (SE 3.2) for women aged 15 and above in Table 2, -2.0 minutes (SE 3.4) for women aged 15 to 64, and 1.9 percentage points (SE 0.9) higher in the usual-status employment rate. All are small, and their signs disagree, so I read them together as showing no consistent association between household care need and women's paid work, and certainly not as evidence that care raises it.

The second and third columns split need by whether the household reports a caregiver available. Where a caregiver is available, women's adult care rises by 37.9 minutes (SE 1.4); where none is available, by 20.8 minutes (SE 2.8). The difference is 17.1 minutes (SE 3.1) and statistically significant at the 1 percent level. The pattern is consistent with availability meaning, in practice, the presence of a woman to whom care can be assigned: where she exists, she provides it, and where she does not, the household reports no caregiver and less care is recorded. For employment, the coefficients are -3.7 minutes (SE 3.6) with a caregiver available and 9.7 minutes (SE 10.9) without (difference -13.3 minutes (SE 11.5)). The estimates for households without an available caregiver rest on 449 women and are correspondingly imprecise.

These averages combine households with very different levels of need. The survey does not record severity, and the diary records one day. As in high-income countries (Lilly et al. 2007; Carmichael and Charles 2003), the burden is likely to be concentrated among a minority of intensive carers whose costs are well above the mean.

6.4 Who provides adult care

Table 4: Distribution of dependent-adult care minutes by carer's sex and relationship to the probable recipient

Carer's relationship to the recipient	Women	Men	Women, recipient 60+	Men, recipient 60+
Spouse	21.4%	7.4%	25.0%	5.4%
Parent	17.5%	3.3%	0.0%	0.0%
Son or daughter	6.2%	10.9%	6.8%	17.2%
Son- or daughter-in-law	15.9%	0.0%	27.3%	0.0%
Other relative or non-relative	7.2%	1.9%	8.6%	1.1%
Grandchild	3.3%	1.5%	5.8%	2.8%
Sibling	1.7%	0.7%	0.0%	0.0%
Parent-in-law	0.6%	0.0%	0.0%	0.0%
Grandparent	0.4%	0.0%	0.0%	0.0%

Notes: Need households with exactly one probable recipient (2,028 households; 1,080 with a recipient aged 60+). Shares of all adult-care minutes in each sample, summing to 100% over the women and men columns. Relationship to the recipient is derived from both members' relationship to the head (for example, the head's wife caring for the head's parent is a daughter-in-law); 'son or daughter' includes a head caring for a parent, who may be a son-in-law. Carers 15+, excluding probable recipients. Survey-weighted.

Table 4 distributes adult-care minutes by the carer's relationship to the person who needs care. The survey does not name the recipient, so I restrict this analysis to the 2,028 need households with exactly one probable recipient and derive each carer's relationship to that person from both members' relationship to the head. Wives caring for husbands provide 21 percent of adult-care minutes and husbands caring for wives 7 percent. Daughters- and sons-in-law, almost all of them daughters-in-law, provide 16 percent, sons and daughters 17 percent (sons 11 percent, daughters 6 percent). Where the recipient is aged 60 or above, daughters-in-law provide 27 percent of care minutes and sons 17 percent. Under patrilineal norms the duty of caring for aging parents is assigned to the son; in these data the daughter-in-law carries out more of it (Lamb 2000; Kandiyoti 1988). The derivation is approximate: the probable-recipient flag is imperfect, and a head caring for a parent-in-law is counted as a child.

6.5 Care over the life course

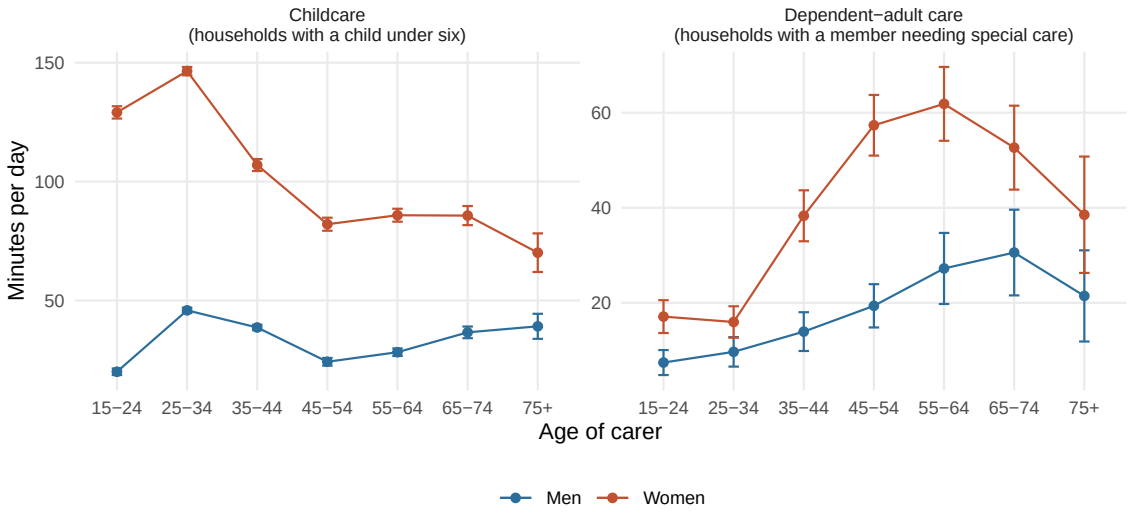


Figure 1: Care by age and sex of the carer, in households with each type of need. Survey-weighted means with 95% confidence intervals; persons aged 15+, excluding probable care recipients.

Figure 1 shows the age profile of care. Childcare in households with a child under six peaks for women aged 25-34, the ages of mothers of young children; at ages 55 to 64, women provide 86 minutes, much of it presumably grandmothers' care of grandchildren. Women's dependent-adult care in households with a special-care need peaks at ages 55-64 and men's at 65-74. People aged 60 and above provide 27 percent of all adult-care minutes in need households (older women 19 percent, older men 8 percent). Among people aged 60 and above in these households, women provide 25.5 minutes (SE 4.4) more adult care than men, and are 22.2 percentage points (SE 2.9) more likely to provide any. The gender gap in adult care does not narrow measurably in old age, unlike the pattern documented for spousal care in high-income countries.

The "sandwich generation" facing both needs at once is a small group. Women aged 35 to 54 facing

both a child under six and a member needing special care are 0.5 percent of women of that age. Their total caregiving is 39.0 minutes (SE 6.8) less than predicted by the two needs separately (statistically significant at the 1 percent level). The negative interaction means that the two needs do not add up: women facing both provide less care than the sum of the two separate associations, consistent with a ceiling on the time one woman can give or with care being shared with other members. Given how small the group is, the estimate says little about the population as a whole.

6.6 Spouses caring for spouses

The probable-recipient flag allows a closer look at spousal care. Among married persons in need households whose linked spouse is a probable recipient (569 wives and 257 husbands), wives provide 66.9 minutes of adult care a day and husbands 53.6, a gap of 13.3 minutes (SE 6.8) (significant only at the 10 percent level), and wives are 15.1 percentage points (SE 4.2) more likely than husbands to provide any (statistically significant at the 1 percent level). Restricted to carers aged 60 and above, the gap is 16.5 minutes (SE 9.5). In the 131 need households consisting only of an older couple, where the only possible household carer is the spouse, the gap is -9.0 minutes (SE 16.1). When it is their spouse who needs care, husbands are less likely than wives to provide any, but the gap in minutes is imprecise and not statistically significant at the 5 percent level, and among older couples living alone it disappears, consistent with the role of men as spousal carers documented in other settings. For older couples living on their own, the question of who cares has only one answer within the household.

6.7 Men's care when no woman is present

Men's care depends on whether a woman is present to provide it. 8.9 percent of households with a special-care need contain no non-recipient woman aged 15 or above. Men in such households (247 in the sample) provide 46.3 minutes (SE 6.5) more dependent-adult care than men in need households with a woman present, and are 33.8 percentage points (SE 3.9) more likely to provide any. With controls, the difference is 39.1 minutes (SE 6.1). They also do 111.6 minutes (SE 9.3) more domestic work. Men in households without a woman provide 58.4 minutes of adult care a day, against 37.1 for women in need households. When there is no woman to assign care to, men's adult care is at or above women's average in need households, so men's low care elsewhere is consistent with care being assigned to women rather than with men's inability to provide it; the comparison rests on a small sample, and men caring alone may also face greater need.

7 Discussion

7.1 Interpretation

Three features of care in Indian households stand out once need is held constant. First, women provide 80 percent of childcare and 76 percent of adult care in the households that need each, so

the gender division of care is extreme for both, and somewhat less so for adults than for children. The population gap in care is not an artefact of men's lower exposure to need. Second, the need for adult care has costs in women's time that are invisible in population averages because only 1.9 percent of households report it. For the women in those households, the cost falls more on free time and sleep than on measured paid work. Third, care follows position. Wives caring for husbands provide 21 percent of adult care in single-recipient households, daughters-in-law 16 percent and sons 11 percent; people aged 60 and above provide 27 percent; and men's care rises when no woman is present.

The results bear on the competing accounts in the conceptual framework. Daughters-in-law provide more of the care of older parents than the sons to whom the obligation is assigned, which comparative advantage alone does not readily explain. Men's care rises when no woman is present, which is consistent with an assignment of care by gender and position in which men's care is a fallback rather than a share (Kandiyoti 1988; Folbre 1994). The caregiver-availability answer points the same way: women's care time rises more where a caregiver is reported available, consistent with that caregiver usually being a woman.

7.2 Limitations

The design is a single cross-section, and the estimates are conditional associations. Households in which someone needs care differ in age structure, income and health in ways the controls capture only partly, and families may arrange co-residence according to who is able to care, which would make the carers observed in need households more care-oriented than other women. The direction of this bias for the employment estimates is ambiguous: women with weaker labor-market attachment may be selected as carers, which overstates the cost of care, while households may avoid assigning care to women with good jobs, which understates the cost for women in general.

The need measure is coarse. It records neither severity nor type nor the identity of the recipient, and it covers any member aged five and above. The probable-recipient flag is an approximation: it misses recipients who work and did not receive care on the diary day, and it includes members who received medical care for minor conditions. The caregiver-availability question is the household's own judgement and may be answered differently by households with similar capacity.

A single diary day understates the share of people who ever provide care and overstates its intensity among those who report it (Frazis and Stewart 2012). Primary-activity minutes understate supervisory and simultaneous care (Folbre 2006; Floro 1995). The responsibility of being on call for a frail relative, which constrains women's options for paid work outside the home, is not measured at all. The estimates should therefore be read as lower bounds on the time constrained by care.

7.3 Policy implications

Long-term care has entered India's policy agenda mainly through the health system and the welfare of older persons (UNFPA and International Institute for Population Sciences 2023). The results suggest that it should also be treated as a question of gender equality and women's employment. Four implications follow, framed by the agenda of recognizing, reducing and redistributing unpaid care (Elson 2017; Addati et al. 2018).

Recognize. The household care-need questions in the Time Use Survey are a valuable instrument. Adding the identity of the recipient, the type of need and an indicator of severity would allow official statistics to report care time by recipient, and to track how care burdens change as the population ages. Official summaries of unpaid work may consider reporting care of dependent adults separately from childcare.

Reduce. Home-based nursing, day-care centers for older and disabled persons, and assistive devices reduce the time that care requires. They should be evaluated not only by their health effects but also by their effects on carers' time and employment, which this survey can measure.

Redistribute. Daughters-in-law provide more of the care of the old than sons do. Men's care rises when no woman is available, so men can and do care when care is not assigned to a woman. Policies that treat the "family" as the provider of long-term care, without specifying who in the family, in practice assign it to women. Support for carers, such as respite services, carer allowances or leave entitlements in formal employment, should be available to men as well as women, and could be designed to encourage sharing.

Prepare for older couples. People who are themselves over 60, often spouses, provide 27 percent of adult care in need households. As fertility falls and migration separates generations, more older couples will live alone. Their care capacity is limited by their own health, and they are the group for whom public provision most directly substitutes for family care.

8 Conclusion

Conditioning on need changes how care in India looks. In households that need care, women provide 80 percent of childcare minutes and 76 percent of dependent-adult care minutes, so the gender gap in care is not a gap in exposure. For working-age women, a member who needs special care is associated with 2.0 minutes (SE 3.4) less time in paid work and 29.0 minutes (SE 2.9) less free time. Adult care is provided mainly by wives and daughters-in-law, and people aged 60 and above provide 27 percent of it. As the population ages, this form of care will grow. Without public provision and a redistribution of care between men and women, most of it will be supplied by women's time.

Declaration of generative AI use

Generative AI tools (Claude, Anthropic) were used to assist with writing and debugging analysis code, checking references, and drafting and editing text. The author reviewed and verified all analyses, results, and text, and takes full responsibility for the content.

Data and code availability

The Time Use Survey 2024 unit-level data are distributed by the National Statistics Office, Ministry of Statistics and Programme Implementation, Government of India (microdata.gov.in, catalogue 236), to registered users. Code available from the author on request; will be made public on acceptance.

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Appendix

Table 5: From the household roster to the analysed diaries

Step	N
Persons aged 6+ on the household roster	454,192
Without diary records in the file	3,735
Diaries (person-days)	450,457
Diaries not summing to exactly 1,440 minutes	5
Third-gender diaries (excluded from women-men comparisons)	49
Diaries of women and men	450,408

Notes: All counts are computed from the microdata at knit time. Persons without diary records appear only on the household roster.

Table 6: Activity categories and ICATUS 2016 codes

Category	ICATUS 2016 codes
Employment	110, 121–129, 131–136, 139, 141–142, 150, 160, 170
Employment-related travel	181–182
Own-use production of goods	211–218, 221–227, 229–230, 241–242, 250
Food and meals	311–314, 319
Cleaning and upkeep	321–325, 329
Care of textiles	341–344, 349
Shopping	371–372
Unpaid domestic services (division 3)	311–314, 319, 321–325, 329, 331–333, 339, 341–344, 349, 351–352, 359, 361–362, 369, 371–372, 380, 390
Childcare	411–417, 419, 442
Care of dependent adults	421–426, 429, 443
Unpaid caregiving (division 4)	411–417, 419, 421–426, 429, 431–432, 439, 441–444, 490
Learning	611–614, 619–620, 630, 640, 690
Free time (divisions 7 and 8)	711–713, 719, 721–723, 729–730, 741–742, 749–750, 790, 811–813, 819, 821–823, 829, 831–832, 841–843, 849–850, 860, 890
Sleep	911–913, 919

Notes: Minutes are summed over major activities in the person-day diary. Codes are three-digit ICATUS 2016 codes as recorded in the survey.

Table 7: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	450,457
Households	139,487
First Stage Units	9,969
Diaries summing to 1,440 minutes	100.0%
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling.

Table 8: Care need in the sample households

Statistic	Value
Households in sample	138,647
Households with a member needing special care	2,641
Share of households with a special-care need	1.9%
Share of special-care households with a caregiver available	84.6%
Share of special-care households with a member 60+	65.6%
Share of special-care households with a probable recipient flagged	89.4%
Share of special-care households with no non-recipient woman 15+	8.9%
Share of households with a child under six	35.1%

Notes: Survey-weighted shares (household weights) except counts. Households with at least one member aged 15+ and non-missing need questions.